PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

code:

PRACTICAL NO 5.py - C:/Users/mvluc/Documents/ds nikita/PRACTICAL NO 5.py (3.14.6)

File Edit Format Run Options Window Help

```
import pandas as pd
data = {
  "name": ["NIKITA", "SAPNA", "RIYA"],
  "age": [70, 80, 90],
  "qualified": [True, False, False]
}
df = pd.DataFrame(data)
newdf = df.filter(items=["name", "age"])
print(newdf)
print("NIKITA")
```

:

Output:

IDLE Shell 3.14.6

File Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 20
MD64)] on win32
Enter "help" below or click "Help" above for m
>>>
========= RESTART: C:/Users/mvluc/Documents/ds
      name  age
0   NIKITA   70
1    SAPNA   80
2     RIYA   90
NIKITA
>>>
```

5b. Write a python code to show statistical information on given data set.

Code:

PRACTICAL NO 5.py - C:/Users/mvluc/Documents/ds nikita/PRACTICAL NO 5.py (3.14.6)

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```
import statistics

print("Mean:")
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mean([1, 3, 5, 7, 9, 11]))
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))

print("\nMedian:")
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))

print("\nMode:")
print(statistics.mode([1, 3, 3, 3, 5, 7, 7, 9, 11]))
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.mode(['red', 'green', 'blue', 'red']))

print("\nStandard Deviation:")
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
print(statistics.stdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.stdev([-11, 5.5, -3.4, 7.1]))
print(statistics.stdev([1, 30, 50, 100]))
```

Output:

```
========= RESTART: C:/Users/mvluc/Documents/ds nikita/P
Mean:
7
6
1.8666666666666667

Median:
7
6.0
1.05

Mode:
3
1
red

Standard Deviation:
3.7416573867739413
0.6925797186365383
8.414471660973927
41.67633221226007
```

5c.Write a python code to create pandas series from dictionaries.

Code:

```
# import the pandas lib as pd
import pandas as pd

# create a dictionary
dictionary = {'DS': 10, 'PDS': 20, 'AM': 30}

# create a series
series = pd.Series(dictionary)

print(series)
```

OUTPUT:

```
========= RESTART: C:/Users/mvl
DS      10
PDS     20
AM      30
dtype: int64
>
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

Code:

PRACTICAL NO 5.py - C:/Users/mvluc/Documents/ds nikita/PRACTICAL NO 5.py (3.14.6)

File Edit Format Run Options Window Help

```
# importing pandas as pd
import pandas as pd

# dictionary of lists
dict = {'name':["nikita", "sapna", "riya", "jasad"],
        'degree': ["bscit", "bsccs", "M.Tech", "MBA"],
        'score':[90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
```

Output:

```
========= RESTART: C:/Users/mvluc/Documents/ds nikita/PRACTICA
         name  degree  score
True   nikita   bscit     90
False   sapna   bsccs     40
True     riya  M.Tech     80
False   jasad     MBA     98
```